Questions

Q1.

An atom contains electrons, neutrons and protons.
(a) Use words from the box to complete the sentences.

neutral	negative
much larger than a neutron	much smaller than a neutron
positive	the same size as a neutron

d

(i) The charge on an electron is .....

(1)

(ii) An electron has a mass that is .....

(1)

(b) At a petrol station, a pipe is used to transfer petrol to the storage tanks.
The pipe is earthed.

There is friction between the petrol and the end of the pipe.

(i) Explain why it is dangerous **not** to earth the pipe.

(2)

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(ii) Explain how earthing the pipe makes this process much safer.

(2)

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*(c) The photographs show some electrostatic effects.

positively charged balloon near hair

positively charged rod near some paper

positively charged balloon near a thin stream of water

Explain in terms of electric charges how one of these effects is caused.
You may include diagrams to help with your answers.

(6)

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(Total for Question is 12 marks)

Q2.

Figure 10 shows a safety sign on the door of a laboratory where radioactive materials are used.

Figure 10

(i) State **one** way that radioactivity can be dangerous to humans.

(1)

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.....

(ii) State **one** piece of equipment that can be used to measure radioactivity.

(1)

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.....

(iii) Alpha (*a*) radiation and ultraviolet (UV) radiation are ionising radiations.
Give **two** other ionising radiations.

(2)

1 .....

.....

2 .....

.....

(Total for question = 4 marks)

Q3.

Americium-241 is a radioactive isotope that emits alpha particles.

Americium-241 is used in smoke alarms.

Give a reason why it is safe to use americium-241 in smoke alarms.

(1)

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.....

(Total for question = 1 mark)

Q4.

Radiotherapy can involve irradiation of patients.

Radioactive tracers can involve contamination of patients.

State **two** differences between irradiation and radioactive contamination.

(2)

1 .....

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2 .....

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(Total for question = 2 marks)

Q5.

Figure 12 is a diagram representing an atom.

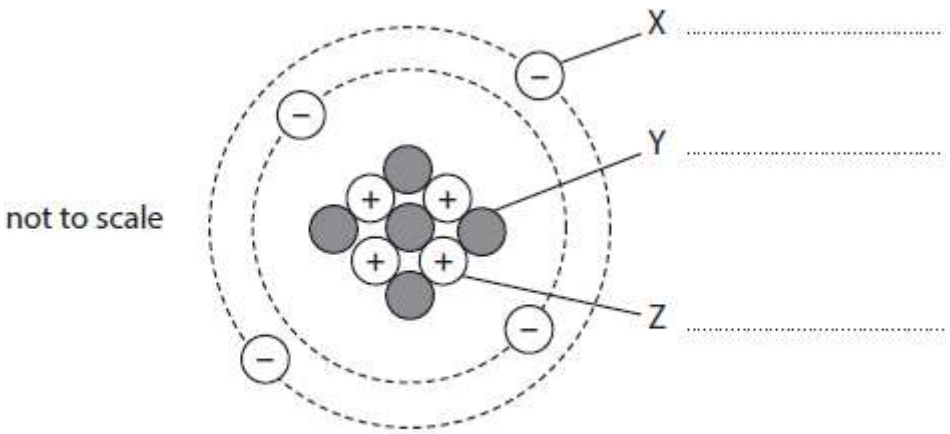

Figure 12

Write the names of the particles X, Y and Z on the lines next to each letter in Figure 12.

(3)

(Total for question = 3 marks)

Q6.

Nuclear power is used for generating electricity.

(i) State **two** advantages of generating electricity using nuclear power compared with generating electricity from gas-fired power stations.

(2)

1 .....

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2 .....

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(ii) Using nuclear power stations to generate electricity is unpopular with many people.
State **two** reasons why nuclear power stations are unpopular.

(2)

1 .....

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2 .....

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(Total for question = 4 marks)

Q7.

Both using nuclear fuel and burning oil produce harmful waste products.

State **one** harmful waste product from each process.

(2)

using nuclear fuel .....

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burning oil .....

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(Total for question = 2 marks)

Q8.

Figure 23 shows the decay curves of two different isotopes, Q and P.

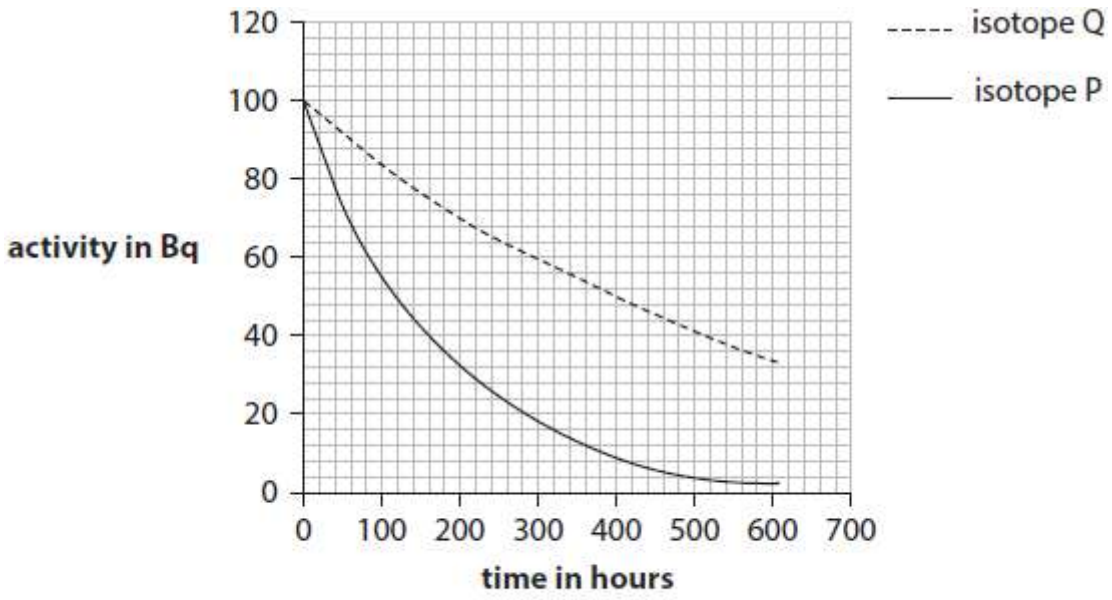

Figure 23

(i) Use the graph in Figure 23 to determine the half-life of isotope P.

(2)

half-life of isotope P = ... hours

(ii) Suggest a reason why the sample of isotope Q could be more dangerous to humans than the sample of isotope P.

(1)

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(Total for question = 3 marks)

Q9.

A teacher demonstrates a radioactivity experiment to a class of students.

The teacher places a radioactive source in front of a radiation detector.

(i) State **one** safety precaution the teacher should take.

(1)

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(ii) The teacher uses the detector to measure the activity of the source several times.
Figure 13 shows the results.

	1st measurement	2nd measurement	3rd measurement	4th measurement
activity in Bq	21	23	19	22

Figure 13

The teacher tells the class that radioactive decay is random.

State how the data in Figure 13 supports this statement.

(1)

.....

.....

(iii) Calculate the mean of the **four** measurements in Figure 13.

(1)

mean = ... Bq

(Total for question = 3 marks)

Q10.

* Nuclear fission is used in nuclear power stations and nuclear fusion takes place in stars.

Describe the similarities and differences between nuclear fission and nuclear fusion.

(6)

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(Total for question = 6 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(a)(i)	negative (1)		**(1)**
(a)(ii)	(much) smaller than a neutron (1)		**(1)**
(b)(i)	An explanation linking • (friction/it) produces charges (at the end of the pipe) (1) • charge jumps to fuel tank (1) • (charge/friction) causes a spark (1) • can cause a fire /explosion (1)	static (electricity) builds up	**(2)**
(b)(ii)	An explanation linking • (excess) charge / electrons (1) • Removed/ conducts away (1)	static charge discharged/ neutralised discharge current scores both marks	**(2)**

		Indicative Content	Mark
QWC	***(c)**	An explanation etc. including some of the following points • static electricity • opposites charges attract • charges are different • induced charges • charges separate • charges move • electrons move • electrons move towards a positive charge / balloon / rod Allow credit for a correct explanation for an effect which is not given in the question. Allow credit for separation of charge being shown on a diagram.	**(6)**
Level	**0**	No rewardable content	

1	**1 - 2**	• a limited explanation. Explains the effect is caused by charges. e.g. the charge on the balloon pulls the water; the charge on the rod attracts the bits of paper; the balloon is rubbed to give it charge; opposites attract; positive and negative attract; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	**3 - 4**	• a simple explanation. Explains an effect is caused by opposite charges attracting or like charges repelling. e.g. the charge on the balloon is opposite to the charge on the water so they attract; the positive charges on the balloon attract negative charges on the girl's hair; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed explanation. Explains the effect is caused by induction, charge separation or moving electrons which leads to attraction between opposite charges. e.g. the electrons have been moved off the balloon so it has a positive charge and attracts the negative charge on the hair; the balloon has a positive charge and induces a negative charge on the stream of water which attracts it; • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Total marks for question = 12

Q2.

	Answer	Additional guidance	Mark
(i)	**One** from: cell damage (1) cancer (1) radiation sickness / poisoning (1) mutation (1) chromosomal damage (1) dna damage (1) skin damage (1) (named) organ damage (1) burns (1) releases ionising radiation (1)	allow ionises / kills cells	(1) **AO1**

	Answer	Additional guidance	Mark
(ii)	**any one** from: Geiger (Muller) (tube/counter) photographic film dosimeter	accept recognisable spellings GM film badge	(1) **AO1**

	Answer	Additional guidance	Mark
(iii)	**any two** from: beta(minus)/β(-) (1) beta + (1) x-rays (1) gamma/γ (1)	accept positron in place of beta + accept proton beam accept electron beam maximum of 1 mark if one incorrect radiation given zero marks if two incorrect radiations given	(2) **AO1**

Q3.

Question number	Answer	Additional guidance	Mark
	alpha cannot penetrate casing	alpha only travel a few cm in air	(1)

Q4.

Question	Answer	Additional guidance	Mark
	any two from irradiation does not make an object radioactive (1) irradiation can be stopped by shielding (1) irradiation source is outside the object (1) irradiation stops when the source is removed (1)	 ORA ORA source of contamination is in or on the object effect of contamination is continuous allow you can move away from irradiation	**2** **AO1.1**

Q5.

Question number	Answer	Additional guidance	Mark
	X: electron (1) Y: neutron (1) Z: proton (1)	accept recognisable spellings. accept Y and Z in the wrong order for 1 mark independent of any mark for X	**3** **AO1.1**

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	Any two advantages from: no CO_2 produced / reduces global warming more energy (per kg) no cross-country pipelines no cross-country pipelines	no harmful waste gases to atmosphere high energy density fuel IGNORE reference to unqualified non-pollution cost renewable efficiency sped of production	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	Any **two** reasons for unpopularity from **mp1** public perception that radioactivity is dangerous **mp2** radiation leaks from plant **mp3** nuclear accidents **mp4** risks of terrorist attacks **mp5** production/storage of nuclear waste **mp6** (nuclear) waste radioactive for a long time		(2)

Q7.

Question number	Answer	Additional guidance	Mark
	using nuclear fuel: radioactive substances (1) burning oil: carbon dioxide (1)	 named radioactive substance / nuclear waste greenhouse gases named pollutant toxic/poisonous gases atmospheric pollutant / acid rain	(2)

Q8.

Question	Answer	Additional guidance	Mark
(i)	correct use of the graph (1) 110 to 130 (hours) inclusive (1)	horizontal line in range 48-52 Bq to line for either isotope allow two horizontal lines showing halving of the activity award full marks for the correct answer without working	**2** **AO2.1**

Question	Answer	Additional guidance	Mark
(ii)	**one** suggestion from: (Q) has longer half-life (1) (Q) does not decay as quickly/ takes longer to decay (1) (Q) (maintains) a higher (level of) activity (1) (Q) gives a higher <u>dose</u> (1) (Q) emits more ionising (type of) radiation (1)	allow reverse arguments activity decreases more slowly (always) more (radio)active / more decays per second	**1** **AO2.2**

Q9.

Question number	Answer	Additional guidance	Mark
(i)	any **one** from: keep a safe distance (1) point the source away from people (1) handle the source with tongs/at a distance (1) limit exposure time/return source to store (asap) (1) use shielding (1) use of gloves / mask (1) protective clothing (1) wear a film badge/monitor (1)	(store in) lead-lined box use of screen PPE ignore goggles	**1** **AO1.2**

Question number	Answer	Additional guidance	Mark
(ii)	the activity varies (slightly) (1)	numbers / counts / results / measurements are different no pattern increases and decreases ignore random(ly)	**1** **AO1.1**

Question number	Answer	Additional guidance	Mark
(iii)	21 (1)	$\frac{21+23+19+22}{4}$ or $\frac{85}{4}$ accept 21.3 or 21.2 or 21.25 do not accept 21.5 (median)	**1** **AO3.2**

Q10.

Question number	Indicative content	Mark
	Answers will be credited according to the candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 6 marks** <u>Nuclear fission</u> • Splitting a nucleus (uranium) releases energy • loss of mass • conversion of mass to energy • produces daughter nuclei • can be controlled • does not occur naturally <u>Nuclear fusion</u> • (two hydrogen) atoms(protons) joining • loss of mass • releases huge amounts of energy • conversion of mass to energy • needs very high temperatures and pressures	**(6)**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of science ideas lacks detail (A01) • Presents an explanation with some structure and coherence (A01)
Level 2	3–4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5–6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)